



(Legacy) Using ^%ETN for Error Logging

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(Legacy) Using ^%ETN for Error Logging

An older style of error logging uses the ^%ETN utility, described here for reference.

The ^%ETN utility logs an exception to the application error log and then exits. You can invoke ^%ETN (or one of its entry points) as a utility:

ObjectScript

```
DO ^%ETN
```

Or you can set the \$ZTRAP special variable equal to ^%ETN (or one of its entry points):

ObjectScript

```
SET $ZTRAP="^%ETN"
```

You can specify ^%ETN or one of its entry points:

- **FORE^%ETN** (foreground) logs an exception to the standard application error log, and then exits with a HALT. This invokes a rollback operation. This is the same operation as ^%ETN.
- **BACK^%ETN** (background) logs an exception to the standard application error log, and then exits with a QUIT. This does not invoke a rollback operation.
- **LOG^%ETN** logs an exception to the standard application error log, and then exits with a QUIT. This does not invoke a rollback operation. The exception can be a standard %Exception.SystemException, or a user-defined exception.

To define an exception, set **\$ZERROR** to a meaningful value prior to calling **LOG^%ETN**; this value will be used as the Error Message field in the log entry. You can also specify a user-defined exception directly into **LOG^%ETN**: `DO LOG^%ETN("This is my custom exception")`; this value will be used as the Error Message field in the log entry. If you set **\$ZERROR** to the null string (`SET $ZERROR=""`) **LOG^%ETN** logs a <LOG ENTRY> error. If you set **\$ZERROR** to <INTERRUPT> (`SET $ZERROR="<INTERRUPT>"`) **LOG^%ETN** logs an <INTERRUPT LOG> error.

LOG^%ETN returns a %List structure with two elements: the \$HOROLOG date and the Error Number.

The following example uses the recommended coding practice of immediately copying **\$ZERROR** into a variable. **LOG^%ETN** returns a %List value:

ObjectScript

```
SET err=$ZERROR
/* error handling code */
SET rtn = $$LOG^%ETN(err)
WRITE "logged error date: ", $LIST(rtn,1), !
WRITE "logged error number: ", $LIST(rtn,2)
```

Calling **LOG^%ETN** or **BACK^%ETN** automatically increases the available process memory, does the work, and then restores the original **\$ZSTORAGE** value. However, if you call **LOG^%ETN** or **BACK^%ETN** following a <STORE> error, restoring the original **\$ZSTORAGE** value might trigger another <STORE> error. For this reason, the system retains the increased available memory when these ^%ETN entry points are invoked for a <STORE> error.

